MILITARY SPECIFICATION

CONTROL ASSEMBLY, GYROSCOPE, ATTITUDE

A/A24G-39

This amendment forms a part of Military Specification MIL-C-38418E dated 30 July 1981, and is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, line 7: Delete “3.6”, substitute “3.7”.

PAGE 2

* 2.1: Add the following specifications:

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-E-17555	Electronic and Electrical Equipment Accessories and Repairs Parts: Packaging and Packing of

PAGE 3

* 2.1: Add the following Standards:

MILITARY

MIL-STD-129	Marking for shipment and storage
MS-90376	Cap, Dust, Plastic, Electric Connectors

FSC 6610

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PAGE 4

* 3.2, Subparagraph a(3): Add the following compass system controllers to the existing list:

or, DCK-243/A24G (moisture sealed, red-lighted)

or, DCK-244/A24G (timed auto sync, white-lighted)

or, DCK-245/A24G (timed auto sync, white-lighted, IPL)

or, DCK-246/A24G (timed auto-sync, red-lighted)

PAGE 6

3.5.3, line 4: Delete “80”, substitldte “90”.

PAGE 7

* 3.5.3: Add the following compass system controllers to the existing list:

k. DCK-243/A24G Compass System controller: 1O,O00 hours

1_{0} DCK-244/A24G Compass System Controller: 10,000 hours

m. DCK-245/A24G Compass System Controller: 10,000 hours

n. DCK-246/424G Compass System Controller: 10,OOO hours

3.6: Delete paragraph and substitute: “The displacement gyroscope shall consist of a vertical and a directional gyro mounted in multiple gimbals and associated electronics as described in the following paragraphs. The gimbal arrangement shall be designed to avoid gimbal lock position. The gimbal assembly shall be heretically sealed. The gyro trim circuitry and the elapsed time indicator shall be external to the sealed assembly (Figure 2).”

PAGE 9

3.6.4.3.3: Add the following sentence to the end of the paragraph:

“Whenever fast. erection is in effect, 78VøC shall be present as an erection warning signal at connector 1J1-4.

PAGE 11

TABLE I: After “ERECTION Warning,” add “(78VøC fast erect).”

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PAGE 11

TABLE I: The function for each PIN 13, 14, and 15 shall be changed as follows: Delete “Pitch output øA excited” and substitute “Roll output øA excited” for each of these three pins.

PAGE 18

3.7.2: Add the following sentence to the end of the paragraph: “For the ASK-44/A24G, at least 30° rotation of the heading set control knob on the compass system controller shall be necessary to develop a drive rate of 30° per second.”

PAGE 21

3.7.6.2, line 2: Delete “fault detected” and substitute “fault is detected”.

PAGE 22

* 3.8, line 2: Change the second sentence to read: “The configuration shall be in accordance with figure 4, 5, or 7.”

* 3.8, Subparagraph c, line 1: Delete "DCK-128/A24G and DCK-234/A24G" and substitute “DCK-128/A24G, DCK-234/A24G, and DCK-244/A24G”.

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* 3.8, Subparagraph d, line 1: Delete “DCK-129/A24G and DCK-235/A24G” and substitute “DCK-129/A24G, DCK-235/A24G, DCK-243/A24G, and DCK-246/A24G”.

* 3.8, Subparagraph e, line 1: Delete “DCK-228/A24G and DCK-236/A24G” and substitute “DCK-228/A24G, DCK-236/A24G, and DCK-245/A24G”.

* 3.8, Subparagraph f, line 2: Delete “DCK-234/A24G, DCk-235/A24G, and DCK-236/A24G” and substitute “DCK-234/A24G, DCK-235/A24G, DCK-236/A24G, and DCK-243/A24G”.

* 3.8.2.1: In the MS25169P2 column, add:

DCK-244/A24G
DCK-245/A24G
DCK-246/A24G

* 3.8.2.1: In the MS21384-16 column, add:

DCK-243/A24G

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* 3.8.3, line 2: Delete “figures 4 and 5“ and substitute “figures 4, 5, and 7“.

* 3.8.3.2: Add the following compass system controllers to the existing list:

DCK-243/A24G
DCK-244/A24G
DCK-245/A24G
DCK-246/A24G

* 3.8.3.3, Subparagraph a: Add the following compass system controllers to the existing list:

DCK-243/A24G
DCK-244/A24G
DCK-245/A24G
DCK-246/A24G

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* 3.8.4.1, line 2: Delete “Sourthern” and substitute “Southern”

PAGE 29

3.17.1.?, line 4: Delete “(dewpoint 65°C)” and substitute “(dewpoint -65°C)”.

PAGE 30

3.20, line 1: Delete “18.56” and substitute “18.65”.

3.20, line 2: Delete “18.71” and substitute “18.8”.

PAGE 42

4.6.2.3, Paragraph a, line 2: Add “4.6.6” after “4.4.1.1”.

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* 4.6.3.7, Subparagraph a: Add the following compass system controllers to the existing list:

(7) DCK-243/A24G 5.0 ± O.1V
(8) PCK-244/A24G 4.5 ± 0.05V
(9) DCK-245/A24G 5.0 ± O.1V
(10) DCK-246/A24G 5.0 ± O.1V

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4.6.4.h: Delete paragraph and substitute: "A single pole, double throw switch, spring-loaded to the normally-closed position to 26VøC shall be provided. The normally open position shall be closed to 78VøC while actuated. The switch output shall be connected to 2J2-4 and both voltages shall be references to 2J2-9 or equivalent ground. "

PAGE 64

4.6.4.11.n, line 2: Delete "4.6.4.g" and substitute "4.6.4.h".

PAGE 65

* 4.6.4.11, Subparagraph n(l): Delete "3.7.6.2" and substitute "3.7.6.3".

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4.6.4.14.2.1.C(4): Delete "2J2-8 to 2J2-9" and substitute "2J2-17 to 2J2-18".

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4.6.5.3.5.c, line 1: Delete the first word "Rotate" and substitute "For the ASK-32/A24G-39 Amplifier, rotate...".

4.6.5.3.5.c, line 2: Add the following sentence between the first and second sentence: "For the ASK-44/A24G Amplifier, rotate the displacement gyroscope through a 10° right turn."

4.6.5.3.5.d, line 1: Delete the first word "Rotate" and substitute "For the ASK-32/A24G-39 Amplifier, rotate...".

4.6.5.3.5.d, line 1: Add the following sentence between the first and second sentence: "For the ASK-44/A24G Amplifier, rotate the azimuth detector through a 20° right turn."

4.6.5.3.6, line 6: Delete "4.5.3.5" and substitute "4.4.1.1, 4.6.6,".

4.6.6, line 11: Delete "20" and substitute "30".

4.6.6, line 12: Delete "9" and substitute "8".

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* 4.6.14, line 2: Delete "that" and substitute "test".

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* Delete paragraphs 5.1 and 5.2 and substitute with:

"5.1 General. Packing shall be in accordance with the general requirements of MIL-E-17555, supplemented as follows:

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order, preservation-packaging shall be as specified in MIL-P-116, Method III.

5.3 Shipping containers. The Displacement Gyro shall be shipped in specially designed containers that will assure protection from environmental and physical damage. External containers shall meet the standards of ASO special contianers 80132 as specified in drawing no. 15345 (NSN 2RH-6145-01-080-6707MF). Before being place in the special container, the unit shall be packed in a fiberboard container conforming to PPP-B-636. Sealing (taping) and reinforcing (boarding) of the liner package shall be in accordance with the appendix to PPP-B-636.

5.4 Marking. In addition to any special marking required by the contract, each unit package, intermediate and exterior container and utilized load shall be marked as specified in MIL-STD-129.

5.5 Protective covers. Electrical connectors shall be capped with clean, durable shipping caps as specified in MS-90376."

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6.2.2, line 12: Delete "UDI-T-21350" and substitute "DI-R-7034".

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Figure 2. NOTES: Add a new note as follows: 3. Access for mounting bolts (4 places) and installation shall be provided at the top and at the ends or sides of the displacement gyroscope.

PAGE 87

Figure 3, Note 1: Delete "7.01" and substitute "7.1".

* Figure 3: Add ±0.04 tolerance to the 5.03 dimension.

PAGE 88

Figure 4, Backview: Delete "5 $^{+0.00}_{-0.01}$" and substitute "5 $^{+0.00}_{-0.10}$"

Figure 4, Sideview: Delete "3.056 ± 0.056" and substitute "3.06 ± 0.15".

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* Figure 4, Add to notes: "3. weight 1.5 lbs max for units with TAS circuit".

* Figure 4: Add to nomenclature list of compass system controllers:

With the TAS circuit:
DCK-244/A24F (white lighted)
DCK-245/A24G (IPL white lighted)
DCK-246/A24G (red lighted)

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Figure 5, Backview: Delete "3.06 ± 0.06" and substitute "3.06 ± 0.15".

* Figure 5, Note 2: Delete "over 0.25 inch ± 0.003" and substitute "over 0.25 inch ± 0.02" and delete "3 place = ± 0.003" and substitute "3 place = ± 0.005".

Figure 6, at Connector 2J3, Pin 20: Delete "R1 COMPXMTR GRD" and substitute "R1 COMPXMTR EXCIT".

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* New Page: Figure 7. MOISTURE SEALED COMPASS SYSTEM CONTROLLER

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Preparing Activity:
NAVY-AS

(Project 6610-N093)

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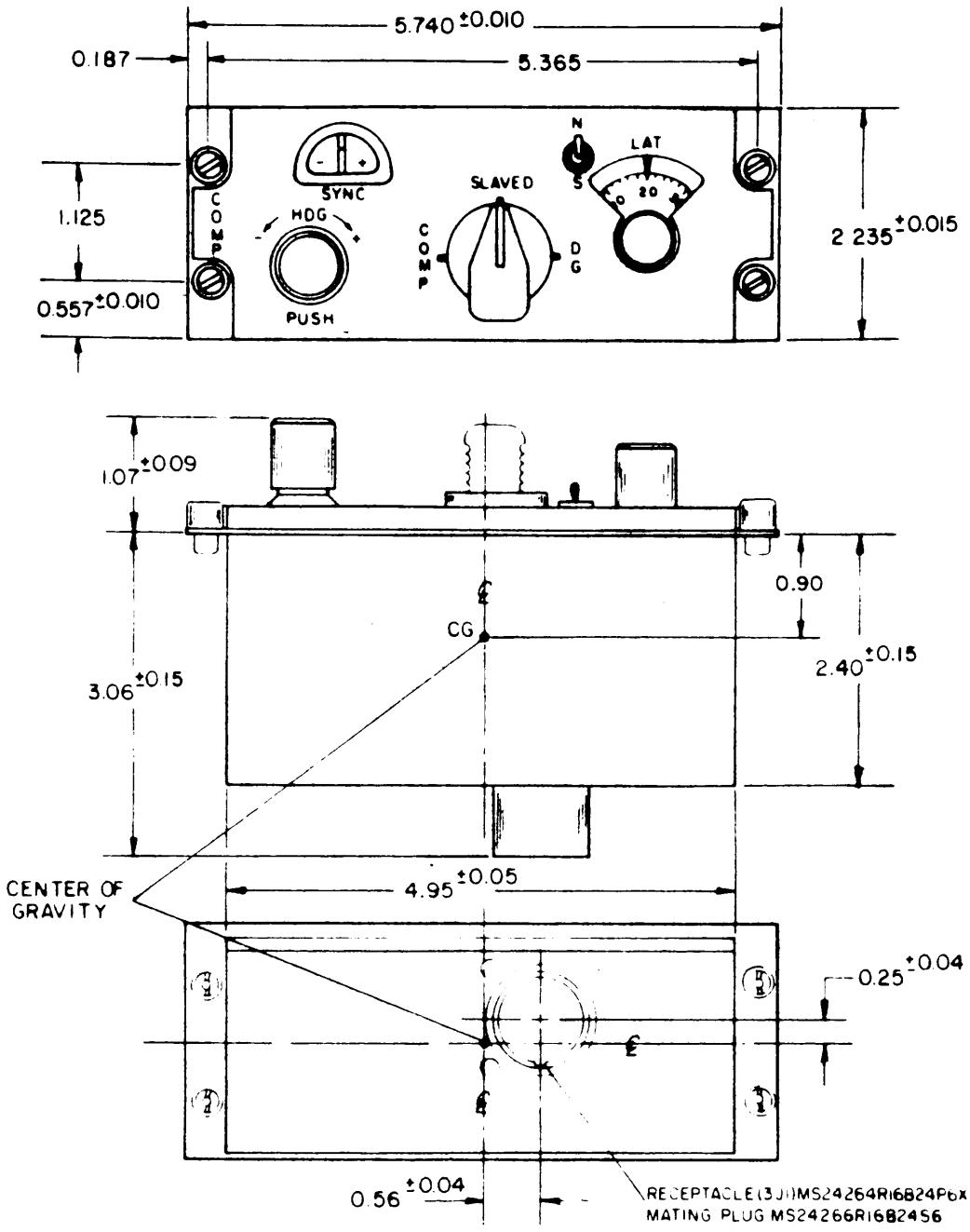

NOTES
1. WEIGHT 1.5 LBS MAX
2. ALL DIMENSIONS ARE IN INCHES
UNLESS OTHERWISE SPECIFIED, DECIMAL TOLERANCES
2 PLACE = 0.25 INCH OR LESS ±0.01, OVER 0.25 INCH ±0.02
3 PLACE = ±0.005
3. CG LOCATION ± 0.25

DCK-243/A24G (RED LIGHTED)

FIGURE 7. MOISTURE SEALED COMPASS CONTROLLER

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